Solutions

LAB LESSON

OBJECTIVE

In this lesson, students will learn to identify the solvent and solute in solutions. They will begin to use words such as diluted, concentrated and concentration.

ACARA CONTENT DESCRIPTIONS

Mixtures, including solutions, contain a combination of pure substances that can be separated using a range of techniques (ACSSU113)

- recognising the differences between pure substances and mixtures and identifying examples of each
- identifying the solvent and solute in solutions

Questioning and Predicting:

Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge (ACSIS124)

- working collaboratively to identify a problem to investigate

Processing and analysing data and information:

Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence (ACSIS130)

- referring to relevant evidence when presenting conclusions drawn from an investigation

LESSON PLAN

Activities	Resources
Activity 1: Homogeneous Solutions vs. Heterogeneous Solutions Start the lesson by posing and discussing the following questions: • What is the difference between homogenous and heterogenous mixtures? • What are some examples of each that you encounter in your daily life? Link the questions to this lesson's topic on solutions, otherwise known as homogeneous mixtures. 5	
Activity 2: The World of Solutions Give out the The World of Solutions worksheet. Note that the answers are found in both the video and the presentation. Use the first 4 slides of the presentation to explain key concepts of solutions, and allow students to fill in the worksheet as much as possible. Play the video and ask students to complete the worksheet. Go through the answers with the students. Open the presentation to slide 5 and have students complete the task as a class. 20	• Photocopies of the The World of Solutions worksheet • Presentation: Solutions • ClickView video *Solutions* **https://clickv.ie./w/lp/8**
Activity 3: Fireworks in a Beaker Give out the Fireworks in a Beaker worksheet. Separate students into groups of 3-4 and read the instructions aloud. Allow approximately 20 minutes for students to complete the experiment and answer the questions. Discuss answers as a class. 20	• Photocopies of the Fireworks in a Beaker worksheet • 50 mL of oil, small beaker (approx. 100 mL), food colouring, toothpick, large beaker (approx. 500 mL), warm tap water
Activity 4: Bingo Mate! Give out the Bingo Mate! worksheet and explain the rules below. Open the presentation to the first slide to provide hints if students cannot think of the answers. Rules of Bingo Mate!: • Students are to approach 9 other students to answer a question each • Whoever gets all of the questions correct wins 15	• Photocopies of the Bingo Mate! worksheet • Presentation: Solutions

ANSWERS

The World of Solutions

1. When a solute  is added into a solvent  , it forms a solution  or homogeneous mixture.

A solute "disappears" in the solvent. If a solute can dissolve, it is said to be soluble and forms a solution. A solute can be a solid, a liquid or a gas.	A solvent does the dissolving of the solute (making it "disappear"), resulting in a uniform composition. A solute can be a solid, a liquid or a gas.	**Properties of Solutions** -Particles spread evenly throughout the solution -Uniform composition (components are all in the same state) -Particles do not settle -Cannot be distinguished by the eye **Concentration** -Measures how much solute is in the solvent Diluted: low concentration of solute Concentrated: high concentration of solute Saturated: no more solute can dissolve

2. a) concentrated
   b) diluted
3. sugar: solute
   water: solvent

Fireworks in a Beaker

1. Suggested Answer:
   When the food colouring was stirred into the oil, the food colouring droplets separated into smaller droplets.
   As oil does not dissolve in water, the oil-food colouring mixture floated on top of the water when it was added.
   The coloured droplets sank to the bottom of the oil and they mixed with the water, causing bursts of colours.
   The food colouring dissolved in water but not in oil.
2. oil, water;
   food colouring, water
3. water; food colouring

Bingo Mate!

Possible Answer:

solvent solute	less	It measures how much solute is in the solution
homogeneous	It dissolves the solute	salt solution
The substance that a solvent can dissolve	A saturated solution	Particles: • have uniform composition • are evenly distributed • do not settle

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The World of Solutions

Use the presentation and the ClickView video to complete this worksheet.

1. Fill in the missing words.

When a s_____  is added into a s_____  , it forms a solution or 

h_____

m_____.

A solute "d_____" in the solvent.

If a solute can dissolve, it is said to be s_____ and forms a s_____.

A solute can be a _____, a _____ or a ___.

A solvent does the dissolving of the solute (making it "disappear"), resulting in a uniform composition.

A solvent can be a _____, a _____ or a ___.

Properties of solutions

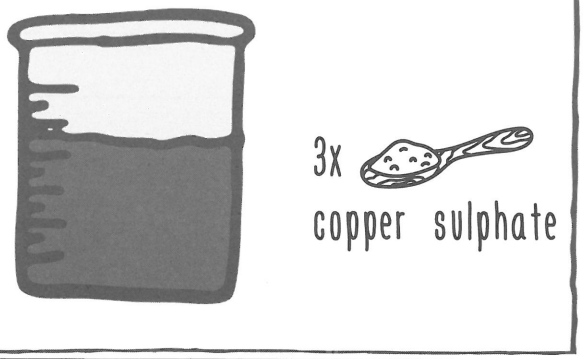
-Particles spread e_____ throughout the solution
-U_____ composition (components are all in the same state)
-Particles do not s_____
-Cannot be distinguished by the e__

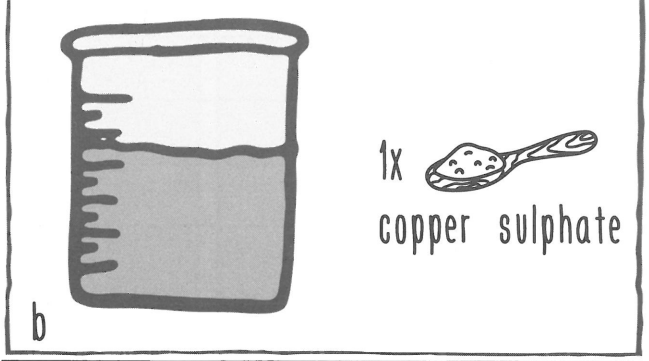
Concentration

-Measures how much s_____ is in the s_____
D_____: low concentration of solute
C_____: high concentration of solute
S_____: no more solute can dissolve

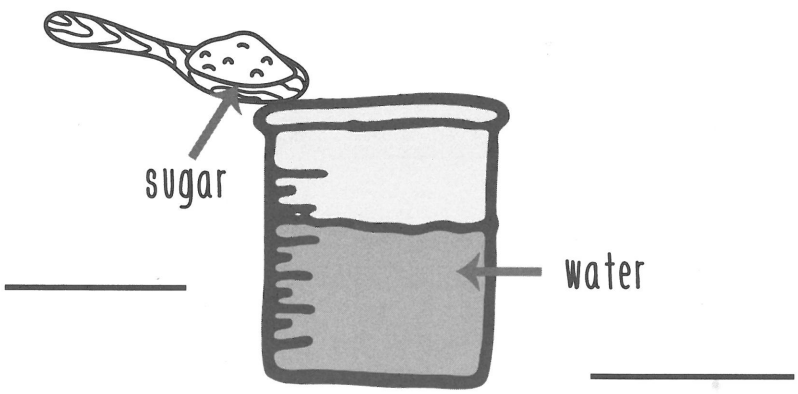
CHEMICAL SCIENCES

2. Label the solutions below as diluted or concentrated.





3. Label the solvent and solute in the diagram below.

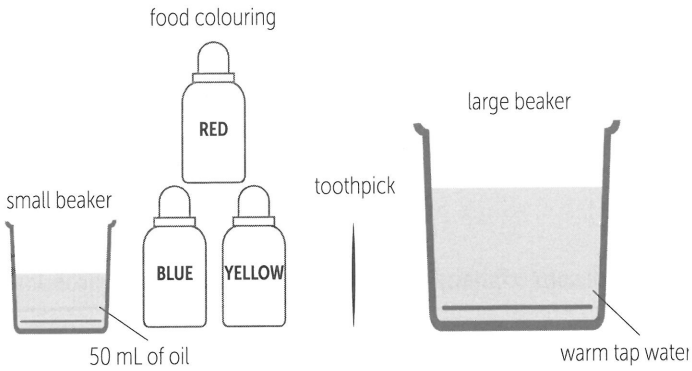


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Fireworks in a Beaker

Materials:

- 50 mL of oil
- small beaker (approx. 100 mL)
- red, blue and yellow food colouring
- toothpick
- large beaker (approx. 500 mL)
- warm tap water



Method:

1. Pour 50 mL of oil into the small beaker.
2. Add three drops of each food colouring into the small beaker.
3. Stir the mixture well with the toothpick.
4. Fill the large beaker ¾ full with warm tap water.
5. Pour the contents of the small beaker into the large beaker.
6. Observe what happens.

Observations

1. Describe what happened in the large beaker.

_____

_____

_____

_____

Questions

2. A heterogeneous mixture and a homogeneous mixture were formed during this experiment. Use words from the box below to describe which two components made up each mixture.

water	oil	food colouring

Heterogeneous mixture: _____ and _____

Homogeneous mixture: _____ and _____

3. What were the solvent and solute in the homogeneous mixture?

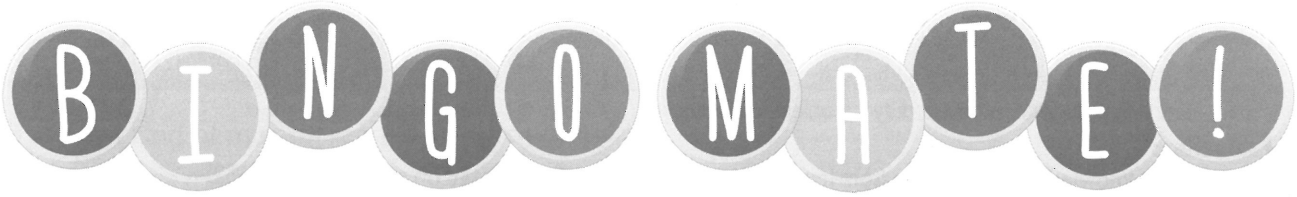
Solvent: _____

Solute: _____

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Bingo Mate!

Instructions:
Approach 9 different classmates and have each of them answer a question below. The first person in the class to get all boxes answered correctly WINS!



Name the components that make up a solution. Answer: Name:	A diluted solution has _____ solute than a concentrated solution. Answer: Name:	What does concentration measure? Answer: Name:
What type of mixture is a solution also known as? Answer: Name:	What does a solvent do? Answer: Name:	Name an example of a solution. Answer: Name:
What is a solute? Answer: Name:	What do you call a solution when it is not able to dissolve any extra solute? Answer: Name:	What is one characteristic of a solution? Answer: Name:

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